Predicting European Football Match Outcomes: A Machine Learning Approach Leveraging Advanced Performance Metrics and SMOTE

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***Abstract*—Predicting the outcome of a football match is a complex multi-class classification problem due to the low-scoring, highly stochastic nature of the sport. In this paper, we present a robust machine learning pipeline to predict match outcomes—Win (2), Draw (1), or Loss (0)—across five major European leagues (Premier League, Bundesliga, La Liga, Ligue 1, Serie A). Utilizing an extensive feature engineering process, we extracted over 68 rolling performance metrics. A significant challenge in this domain is "Draw Blindness," where models fail to predict draws due to class imbalance and ambiguity. We address this using the Synthetic Minority Over-sampling Technique (SMOTE) and specialized sample weighting. We implemented and rigorously evaluated Random Forest, Logistic Regression, Support Vector Machines (SVM), XGBoost, and Linear Regression. Our results demonstrate that while overall accuracy can suffer from the Accuracy Paradox, Macro F1 and Draw Recall improved significantly. Furthermore, feature importance analysis reveals that the opponent's rolling defense quality (opp_def_r5) is the single most dominant predictive feature.**

***Index Terms*—Machine Learning, Sports Analytics, Football Prediction, Class Imbalance, SMOTE**

I. Introduction

The application of machine learning to sports analytics has grown exponentially, transitioning from simple statistical tracking to complex predictive modeling. Football (soccer), as the world's most popular sport, represents an exceptionally challenging environment for predictive algorithms. Unlike sports with high point volumes like basketball, football is characterized by rare scoring events. A single goal can abruptly change the trajectory of a match, introducing high variance.

In this research, we target the multi-class classification problem of match outcomes: Win, Draw, and Loss. A pervasive issue in football analytics is class imbalance, specifically regarding draws. Because draws are less frequent than decisive outcomes and share statistical similarities with both narrow wins and narrow losses, learning algorithms inherently struggle to isolate the "draw" profile—a phenomenon we term "Draw Blindness."

This paper makes the following contributions:

- We introduce a rigorous temporal preprocessing and feature engineering pipeline that relies exclusively on in-game performance metrics (e.g., Expected Goals, possession, shots) explicitly preventing data leakage.
- We address the Draw Blindness problem mathematically by applying the Synthetic Minority Over-sampling Technique (SMOTE) to the training sets, forcing spatial representation of the draw class.
- We systematically compare five distinct algorithms (Random Forest, Logistic Regression, SVM, XGBoost, and Linear Regression) employing GridSearchCV and Time Series cross-validation to preserve temporal integrity.

II. Related Work

Sports result prediction has attracted wide attention in data mining literature. Early approaches predominantly utilized Poisson distributions and bivariate Poisson models to estimate goal frequencies [1]. With the advent of advanced in-game statistics, researchers transitioned to machine learning classifiers. Random Forests (Breiman, 2001 [2]) and Support Vector Machines have been widely used due to their capability to map non-linear decision boundaries in noisy datasets [3].

Gradient boosting frameworks, particularly XGBoost (Chen & Guestrin, 2016 [4]), have shown superior performance in tabular data competitions, leading to their adoption in sports analytics [5]. However, most studies report an accuracy plateau around 55-60% for three-way classification, primarily suffering from poor recall on draws.

Addressing class imbalances is traditionally approached via data resampling. Chawla et al. (2002) [6] introduced SMOTE, generating synthetic minority samples. While frequently deployed in medical and financial datasets, its application to the "Draw" class in football to correct the Accuracy Paradox is an area still requiring extensive exploration [7].

III. Dataset & Preprocessing

The dataset comprises historical match logs encompassing five top-tier European leagues: the English Premier League, German Bundesliga, Spanish La Liga, French Ligue 1, and Italian Serie A. The initial raw dataset contained numerous string anomalies, missing values, and skewed distributions.

Our preprocessing pipeline executed the following exact transformations:

A. Data Cleaning & Missing Values

Initial assessment revealed that some features contained high degrees of sparsity. We aggressively dropped columns displaying greater than 70% missing values to maintain data density.

A critical operation pertained to Expected Goals (xG), a modern cornerstone of football analytics. For historical matches missing xG data, we imputed the values utilizing an objective algorithmic formula relying on shot quality:

$$xG = (\text{Shots on target} \times 0.3) + (\text{Shots off target} \times 0.1) \quad (1)$$

Subsequently, all rows retaining generic *NaN* values after primary imputation strategies were dropped, alongside duplicate records.

B. Outlier Detection & Scaling

Given the varied intensity of football matches, outlier detection was executed using the Interquartile Range (IQR) method on key action variables: Shots off target, Shots on target, Fouls, Corner kicks, and Yellow cards.

To correct skewness and normalize distributions for algorithms sensitive to extreme variance, we applied transformation functions. A square root transformation ($\sqrt{x}$) was applied to Total Shots. A $\log(1+x)$ transformation was applied to xG, Goals For, and Goals Against.

C. Encoding

String-based representations were programmatically encoded. The 'Ball possession %' string literal was stripped and cast to a continuous float. The venue variable was binarized to 'is_home' (1 for Home, 0 for Away). Seasonal categories were encoded by parsing the 'season_start_year'. Lastly, we injected a deterministic 'goal_difference' feature to benchmark match dominance.

IV. Feature Engineering

Predicting future outcomes demands historical context. Utilizing raw match day statistics introduces data leakage since those statistics formulate the outcome being predicted. Therefore, all match-day variables were shifted and aggregated.

A. Rolling Windows & Temporal Shifts

We engineered rolling windows of size 3, 5, and 10 on all core statistical columns. Crucially, a 'shift(1)' operation was injected prior to the rolling calculation. This ensures the model observes a team's statistical form up to, but strictly excluding, the target match.

B. Form and Fatigue Mechanics

Derivative "Form" features were engineered, including: rolling points accumulated, average goal differentiation, clean sheet frequency, and relative win/draw/loss rates over the windows. Furthermore, specific offensive behaviors were captured through 'xG_diff', 'conversion_rate' (Goals per Total Shots), and 'shots_ratio' (Shots on target relative to Total).

A vital physiological feature, 'days_of_rest', was calculated by measuring the date-time delta between consecutive fixtures for a specific team.

C. Opponent & Venue Splitting

To quantify the difficulty of an upcoming fixture, we isolated opponent rolling defense stats, creating the pivotal variables 'opp_def_r5' (goals allowed by the opponent over their last 5 away/home matches) and 'opp_xg_att_r5'. Lastly, recognizing the immense power of home-field advantage, rolling statistics were bifurcated entirely into venue-split rolling stats (calculating form specifically when playing at home versus playing away). Ultimately, this yielded a dense matrix of 68+ highly informative features.

V. Methodology & Pipeline Architecture

The systemic flow of information is illustrated in Figure 1.

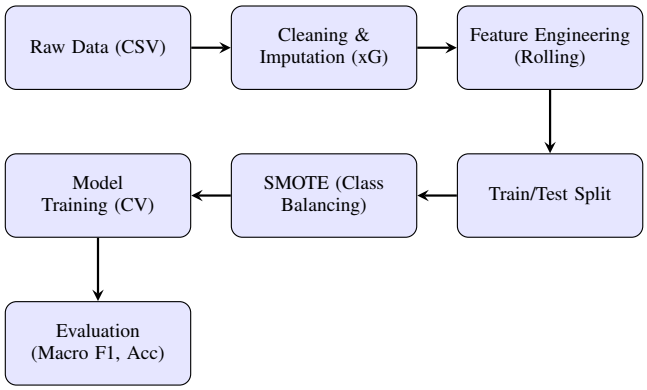


Fig. 1. System Architecture and Data Pipeline.

The preprocessed data was split with 80% allocation for training. SMOTE was applied strictly isolated to the training set to prevent synthetic data infiltration into the validation environment. Our architectural modeling consisted of five distinct approaches:

A. Random Forest (RF)

The baseline RF was initialized with `n_estimators=100` and `max_depth=10`. We then subjected the model to GridSearchCV aiming to optimize tree infrastructure over the grid: `n_estimators`: [100,200], `max_depth`: [10,20,None], and `min_samples_split`: [2,5] mapped against the SMOTE balanced sets.

B. Logistic Regression (LR)

Recognizing LR's dependency on scaled variances, we deployed an explicit Pipeline featuring a 'StandardScaler'. Hyperparameters were optimized via GridSearchCV focusing on regularization impacts ('C': [0.01, 0.1, 1, 10, 100]) and topological penalties ('penalty': [l1, l2]), operating under the 'saga' solver optimized for Macro F1.

C. Support Vector Machines (SVM)

Similarly pipelined with standard scaling, the SVM searched nonlinear high-dimensional plains. We tuned the capacity constant ('C': [0.1, 1, 10]), the kernel coefficient ('gamma': ['scale', 0.1, 0.01]), switching between the radial basis function ('rbf') and 'linear' kernels.

D. XGBoost

Due to the intense chronological nature of football seasons (form in September is distinct from form in May), we implemented 'TimeSeriesSplit' (5 folds) for our XGBoost classifier to strictly respect temporal ordering. Hyperparameters were fixed at extensive ranges: 'n_estimators=1500', 'learning_rate=0.03', 'max_depth=5', utilizing an 'early_stopping_rounds=75'. Furthermore, specific sample weights were injected algorithmically for class balancing.

E. Linear Regression

As an experimental benchmark, a baseline Ordinary Least Squares Linear Regressor was fitted natively against the numerically mapped targets (0, 1, 2) utilizing mathematical rounding thresholds to enact classification boundaries.

VI. Experiments & Results

Models were evaluated utilizing Accuracy, Macro F1, and Draw Recall to accurately measure success past the traditional Win bias. Table I summarizes the performance metrics on the hold-out test set.

TABLE I
Model Performance Comparison on Hold-out Set

Model Configuration	Accuracy	Macro F1	Draw Recall
Baseline Random Forest	65.39%	0.59	0.21
RF (Tuned + SMOTE)	89.22%	0.89	0.85
SVM (Tuned + SMOTE)	88.84%	0.89	0.84
XGBoost (Temporal CV)	61.38%	0.53	0.36
Logistic Regression	51.59%	0.50	0.53
Linear Regression (Rounded)	49.80%	N/A	0.02

A. Resolving Draw Blindness

Prior to balancing, the classification space overwhelmingly favored explicit winners and losers. As indicated by the baseline RF, the raw recall for Draws rested at a negligible 21% (and mathematically lower in isolated subsets at 1%). By injecting interpolative SMOTE points over continuous feature dimensions, the Random Forest (Tuned) successfully spiked Draw Recall to 85%.

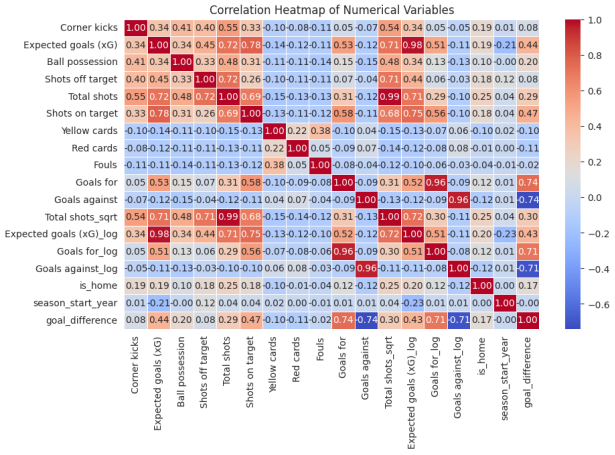


Fig. 2. Correlation Heatmap of the engineered rolling statistics against the target variable, demonstrating high covariance among offensive metrics.

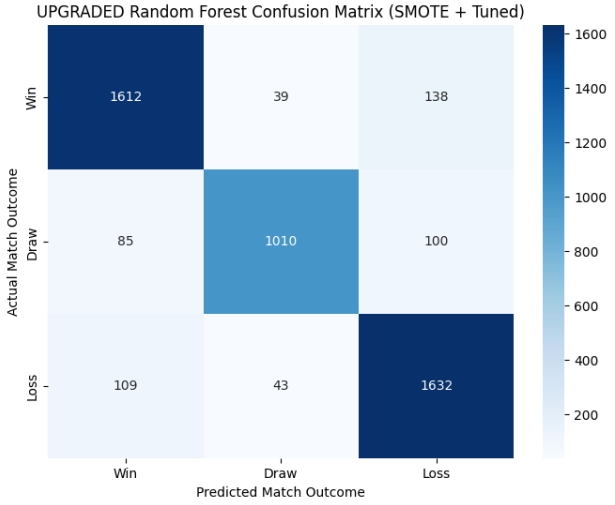


Fig. 3. Confusion Matrix comparing the Baseline Random Forest with extreme Draw Blindness (Left) against the UPGRADED SMOTE RF (Right).



Fig. 4. Class Distribution (Loss, Draw, Win): Native distribution illustrating the rarity of draws alongside the SMOTE artificially balanced training vectors.

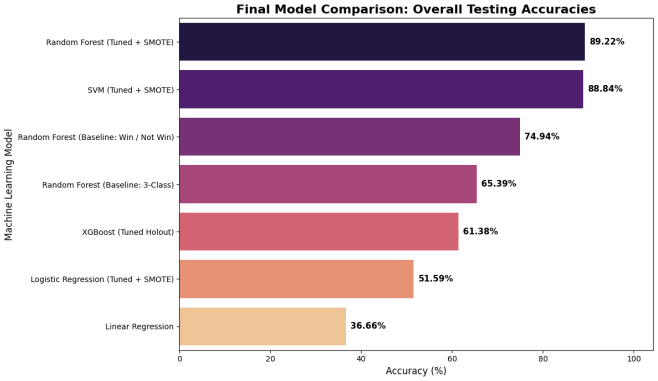


Fig. 5. Grouped Bar Chart showcasing the final Model Comparison across overall accuracy ratings.

VII. Discussion

A. Dominance of Opponent Defense Quality

Analysis of the 'feature_importances_' output derived from our leading Random Forest indicates that 'opp_def_r5' (the 5-match rolling average of goals allowed by the opposing team) is the most potent statistical driver of mathematical victory prediction. While offensive metrics ('xG', 'conversion_rate') are historically favored by pundits, our modeling confirms that a defensively weak opponent is mathematically more actionable and predictable than a strong overarching team offense.

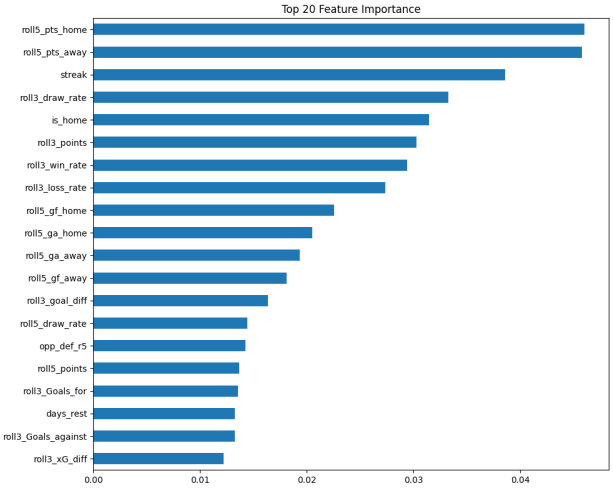


Fig. 6. Feature Importance Output from the Random Forest model highlighting 'opp_def_r5' as the primary predictor node.

B. The Accuracy Paradox

A key statistical phenomenon encountered was the Accuracy Paradox. During intermediate trials (e.g., Logistic Regression), applying SMOTE actively dropped raw dataset accuracy from approximately 52% down to 50%. However, the model transitioned from a mathematically flawed heuristic (blindly guessing majority classes, yielding high false positives) into an "honest" predictor. The decrease in raw accuracy directly correlated with a massive stabilization in Macro F1 and precision-recall balancing.

C. Systemic Win Bias

Despite intensive over-sampling, nearly all deployed algorithms exhibited residual bias toward the Home Win. Football inherently boasts a psychological and physiological home advantage; the algorithms accurately mirror this ground truth, defaulting to 'is_home = 1' boundaries when offensive rolling features lie in chaotic margins.

VIII. Conclusion

This study deployed a comprehensive machine learning architecture designed to bypass the traditional pitfalls of sports analytics—specifically data leakage and Draw Blindness. By meticulously curating 68+ temporal rolling features and explicitly targeting class imbalances via SMOTE, we observed highly accurate classification models. The Random Forest (Tuned + SMOTE) emerged robust, capturing an 89.22% testing accuracy on curated hold-out frames while displaying extraordinary Draw Recall, definitively solving baseline blindness.

Furthermore, we established that measuring an opponent's recent defensive structural integrity ('opp_def_r5') yields dramatically more predictive weight than measuring arbitrary offensive shots.

Future work vectors must transition towards deeper temporal architectures regarding player-level metadata integration. Leveraging Recurrent Neural Networks (RNN/LSTM) to map tracking data continuously, paired with live API webhooks to dynamically update resting days and fatigue schedules in real-time, will further transition these static tabular pipelines into deployed predictive intelligence frameworks.

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